

Targeting Co-Sleeping in Children With Anxiety Disorders Using a Modified Bedtime Pass Intervention: A Case Series Using a Changing Criterion Design

Ariel Ravid

Elizabeth Lagbas

Madeline Johnson

Travis L. Osborne

Evidence Based Treatment Centers of Seattle

Children with anxiety disorders experience high rates of sleep-related problems, with co-sleeping and resistance to sleeping independently being among the more frequent problems reported. Although extinction-based behavioral sleep interventions have repeatedly been shown to be highly effective for treating bedtime resistance, the primary obstacle to their implementation is parent discomfort with these procedures. The bedtime pass intervention was developed to minimize extinction bursts when implementing extinction procedures for childhood sleep problems. Several studies have found this intervention to be effective for treating bedtime resistance behaviors, but not co-sleeping specifically, in nonclinical samples of children. The current paper describes the use of a modified bedtime pass procedure to target problematic co-sleeping and related bedtime resistance behaviors in two children with anxiety disorders who presented for treatment at an outpatient pediatric anxiety specialty clinic. A changing criterion, single subject methodology was used to evaluate the effectiveness of this procedure. Data indicate that both children were able to transition from co-sleeping with parents every night, to sleeping independently, with relatively limited need for

contact with parents at night during the intervention. These findings extend the data for the bedtime pass procedure to both co-sleeping and children with anxiety disorders. Clinical implications of these findings are discussed given the limited guidance for treating comorbid sleep problems in anxious children. Strengths and limitations of the data being drawn from a clinical treatment setting are also discussed.

Keywords: child anxiety; sleep-related problems; co-sleeping; practice-based research

A GROWING BODY OF LITERATURE indicates that children and adolescents with anxiety disorders experience high rates of sleep-related problems (SRPs; see [Peterman, Carper, & Kendall, 2015](#), for a comprehensive review). Multiple studies have replicated the finding that over 85% of anxious youth experience at least one SRP ([Alfano, Ginsburg, & Kingery, 2007](#); [Alfano, Pina, Zerr, & Villalta, 2010](#); [Chase & Pincus, 2011](#); [Storch et al., 2008](#)), with smaller but notable percentages experiencing three or more (55%; [Alfano et al., 2007](#)) and five or more (27%; [Storch et al., 2008](#)) SRPs. Moreover, both anxiety symptom severity ([Alfano et al., 2007](#); [Storch et al., 2008](#)) and number of anxiety diagnoses ([Chase & Pincus, 2011](#)) have been shown to be associated with greater numbers of SRPs.

SRPs refer to a broad range of problems with sleep onset, timing, duration, or quality. These can include, but are not limited to, insomnia, nightmares, sleep-related breathing problems, parasomnias, waking and inability to fall back asleep, and sleeping too little or too much ([Alfano et al., 2007](#)). A specific set of SRPs related to nighttime independence—reluctance/refusal

We would like to acknowledge Maria Hanelin, B.A., and Stacy Shaw Welch, Ph.D., for their contributions to a poster presentation about these data. Some of the data for this manuscript were previously presented at the Association for Behavior and Cognitive Therapies (ABCT) Annual Convention in New York, NY (October 2016).

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Address correspondence to Ariel Ravid, Ph.D., Evidence Based Treatment Centers of Seattle, 1200 5th Ave., Suite 800, Seattle, WA 98101; e-mail: aravid@ebtseattle.com.

to sleep alone and co-sleeping—have repeatedly been shown to be highly prevalent among anxious children and adolescents. [Alfano and colleagues \(2007\)](#) observed that 48% of a sample of youth with generalized anxiety disorder (GAD), separation anxiety disorder, and/or social phobia experienced reluctance/refusal to sleep alone, and [Storch et al. \(2008\)](#) found that 50% of a sample of children and adolescents with obsessive-compulsive disorder (OCD) engaged in co-sleeping. Moreover, these behaviors were the third most frequently reported sleep problems in both samples (out of eight behaviors assessed). A separate study comparing children with GAD to nonanxious controls found that rates of co-sleeping were significantly higher among those with anxiety (36% of children with GAD versus 10.6% of controls) ([Palmer, Clementi, Meers, & Alfano, 2018](#)). Among children with GAD, anxiety severity was significantly associated with a greater prevalence of co-sleeping (as with all SRPs broadly).

Data on the accommodation behaviors of parents of anxious children and adolescents provide further support that co-sleeping is common. A study of parents whose children were presenting at an anxiety specialty clinic found that of the 20 accommodation behaviors examined, allowing children to sleep in the parents' bed was among the top five most endorsed behaviors for both mothers (28%) and fathers (32%) ([Thompson-Hollands, Kerns, Pincus, & Comer, 2014](#)). Moreover, parents also endorsed high rates of sleeping in the child's bed with the child (22% for mothers and 27% for fathers). Interestingly, parents also rated children sleeping in the parents' bed as one of the most interfering of the accommodation behaviors assessed, suggesting that although parents allow this behavior, they may also be quite bothered by it.

The reality that anxious children have higher rates of co-sleeping is not surprising given that co-sleeping is a common way that parents manage a range of behaviors related to bedtime resistance ([Friman et al., 1999](#)). It is also understandable that children with anxiety might struggle more with settling at bedtime given the range of symptoms associated with specific anxiety-related disorders that could interfere with getting to bed and falling asleep ([Chorney, Detweiler, Morris, & Kuhn, 2008](#)). Examples include intrusive thoughts and nighttime rituals associated with OCD, worry thoughts related to GAD, difficulties being alone at night as seen in separation anxiety disorder, nightmares associated with posttraumatic stress disorder, nocturnal panic symptoms that can occur in panic disorder, anxious reviewing or rehearsal of social interactions in social anxiety disorder, and elevated fear associated with a range of phobias.

Considering the numerous ways that anxiety symptoms can potentially interfere with children's sleep, there is a clear need to attend to SRPs when treating pediatric anxiety problems.

Several studies have shown that treating anxiety symptoms in youth can improve SRPs broadly, as well as behaviors specifically related to sleeping independently. [Storch et al. \(2008\)](#) found that children and adolescents who received family-based cognitive behavioral therapy (CBT) for OCD showed significant decreases from pre- to posttreatment in both the total number of SRPs (means of 3.29 and 2.29 respectively) and in co-sleeping specifically (50% and 38.8%, respectively). Similarly, [Alfano et al. \(2007\)](#) found that youth with GAD, separation anxiety disorder, and/or social phobia treated with fluvoxamine displayed significant reductions pre- to posttreatment in both total SRPs (means of 1.4 and 0.61 respectively) and in reluctance/refusal to sleep alone (53.3% and 27.3% respectively). Despite these positive findings, more than 25% of those in each study exhibited problems sleeping independently after anxiety treatment. This is consistent with results from [Peterman et al. \(2016\)](#), who evaluated the impact of CBT for a range of pediatric anxiety disorders on SRPs. Although responders showed a significant reduction in the severity of parent-reported sleep problems, these problems remained clinically significant at posttreatment. These findings suggest that targeting pediatric anxiety symptoms alone may not be sufficient to improve comorbid sleep problems.

To address this issue, [Clementi and Alfano \(2014\)](#) developed a behavioral treatment for children with GAD (Targeted Behavioral Therapy) specifically designed to treat GAD as well as SRPs (including bedtime resistance and co-sleeping). Two of the 14 treatment sessions focused on idiographic assessment and targeting of sleep problems, and additional work on SRPs could occur in other sessions as needed. The treatment was evaluated in a case series of four children with GAD and findings indicated that none of the children met criteria for GAD at 3-month follow-up. Unfortunately, the data on changes in sleep problems were not as promising; at 3-month follow-up only two of the four children met established criteria for clinically significant change on the child-report sleep measure and none met these criteria on the parent-report sleep measure. The authors hypothesized that the sleep portion of the treatment was inadequate to produce lasting sleep-related changes. These data further suggest that ameliorating sleep problems in anxious children will likely require systematic treatment of SRPs to be effective.

Fortunately, effective treatments for a range of childhood sleep problems exist and decades of research indicate that extinction-based methods have the most empirical support in targeting co-sleeping and bedtime refusal (Etherton, Blunden, & Hauck, 2016; Mindell, Kuhn, Lewin, Meltzer, & Sadeh, 2006). Unmodified extinction (or the “cry it out” method) involves parents putting children to bed at a specific time and ignoring all behaviors (e.g., crying, calling out for parents, tantrums) until a specific wake-up time, aside from behaviors related to illness or injury. In graduated extinction, parents follow the same guidelines for ignoring behaviors, with the exception that they check in on the child at fixed or incrementally longer periods of time throughout the night instead of the next morning. Frequently referred to as “sleep training,” parents are instructed to keep check-ins brief (less than a minute) and to provide as little reinforcement of attention seeking efforts by the child as possible. Finally, extinction with parent presence involves the same procedures as unmodified extinction (ignoring all behaviors), but with parents remaining in the room until the child falls asleep (also known as “camping out”).

Despite the effectiveness of extinction procedures for child sleep problems, a major drawback of these interventions is that parents often find them challenging to implement and tolerate. Etherton and colleagues (2016) describe a range of obstacles that parents can encounter when trying to implement extinction-based behavioral sleep interventions, including difficulties with tolerating crying, fears about how extended crying and distress will impact the child and the parent-child bond, methods being inconsistent with personal and cultural beliefs and customs, and practical considerations (e.g., a parent or other child needing to get to sleep), among others. Additionally, all extinction procedures have the risk of leading to temporary extinction bursts (Lerman, Iwata, & Wallace, 1999), in which problematic behaviors increase when reinforcement is removed. Not surprisingly, parent resistance to these procedures remains the top reason extinction-based methods for childhood sleep problems are not used (Etherton et al., 2016).

An alternative to the aforementioned extinction procedures for addressing bedtime problems and resistance (e.g., crying out, leaving the bedroom) is the bedtime pass procedure. First described by Friman et al. (1999) in a case series with two typically developing boys (ages 3 and 10), the procedure involves extinction in that from the start of intervention children do not have unlimited access to parents at night. Instead, children receive one

“bedtime pass” per night that they can exchange for one departure from their bedrooms. Departures are expected to be brief and have a specific purpose (e.g., getting a drink, going to the bathroom, brief contact with parents in the form of a hug or other comforting behavior). Following this departure, children give the pass to their parents and are returned to their bedrooms with minimal attention and parents are instructed to ignore all further bedtime resistance behaviors for that night (i.e., extinction). The procedure then repeats the following night. The purpose of the bedtime pass is to help minimize extinction bursts by using functional communication training and stimulus control to give children a method for appropriately requesting attention and teaching limits around when they can request the attention. An ABAB design was used and rates of problematic bedtime behaviors decreased for both children during the intervention phases as compared to the baseline phases. Bedtime resistance behaviors were reduced to zero by posttreatment (about 5 weeks) for both children and improvements were maintained at 3-week follow-up.

Two additional studies have provided support for the bedtime pass intervention, including a case series with four typically developing 3-year-old boys (Freeman, 2006) and a small randomized control trial with 19 typically developing boys and girls between the ages of 3 and 6 (Moore, Friman, Fruzzetti, & MacAleese, 2007). Results of the case series indicated that all bedtime resistance behaviors were reduced to zero by the end of the intervention without any extinction bursts. Participants in the randomized controlled trial who were assigned to the bedtime pass showed significantly greater reductions in all bedtime resistance behaviors assessed than those in the control condition, and gains were maintained after 3 months. An impressive 93% of those in the bedtime pass condition had near zero rates of leaving the room by the end of treatment (as compared to 44% in the control condition), with 83% meeting this criterion at 3-month follow-up. Importantly, treatment satisfaction for parents was very high in the bedtime pass condition, with 100% of parents reporting that they did not believe their child experienced discomfort due to the intervention and 92% reporting they themselves did not experience discomfort due to the intervention. These data are consistent with those from Friman et al. (1999), who surveyed 20 parents of elementary school children about three methods for responding to bedtime problems and resistance: ignoring (extinction), letting children sleep with parents, and the bedtime pass method. Parents rated the bedtime pass method as significantly more acceptable than

both ignoring (extinction) and co-sleeping, which were rated similarly.

To date, published studies on the bedtime pass procedure have only included children from non-clinical populations and have targeted bedtime resistance behaviors, but not co-sleeping specifically. The purpose of the current paper is to describe the use of a modified bedtime pass intervention in the treatment of two children with anxiety disorders to target co-sleeping and related bedtime resistance. The structure of this paper is guided by the single-case reporting guidelines for behavioral interventions (SCRIBE) when applicable, in order to help standardize the information provided in such reports (Tate et al., 2016).

An important clarification is that in both cases presented, the co-sleeping behaviors were experienced as markedly problematic and interfering by the children's parents. This is to be differentiated from co-sleeping that is culturally normative, the result of life circumstances (i.e., a home in which there are not enough beds for children to sleep separately from their parents or family members), or other contexts in which parents are not bothered by children engaging in co-sleeping (Mileva-Seitz, Bakermans-Kranenburg, Battaini, & Luijk, 2017).

Material and Methods

PARTICIPANTS AND SETTING

Participants were two children and their parents who presented for treatment at an outpatient pediatric anxiety specialty clinic in the Pacific Northwest. The intervention was selected based on whether a primary presenting concern included parent- or child-expressed difficulties with co-sleeping and sleeping independently. At the time treatment was provided, neither the therapist nor the families had explicit intention that the data from treatment would be used for research purposes. That decision was made after treatment concluded in both cases.

Treatment was provided in an outpatient clinic setting. Sessions ranged from 45–60 minutes with children and one or both parents participating in the sessions (except for parent sessions when only parents participated). Materials (bedtime passes) were created in session and taken home. The clients' parents implemented the bedtime pass intervention at home, tracked outcomes, and reported back in the following session. External fidelity checks on the implementation of the intervention were not feasible given the naturalistic setting; however, children and parents agreed 100% of the time on the report of the target behaviors. The therapist was a doctoral-level, licensed clinical psychologist (first author) with specialty training in the treatment of

children with anxiety disorders, as well as childhood sleep problems.

EQUIPMENT

Bedtime passes were idiographically designed during baseline sessions with children and their parents. Symbols or pictures that were meaningful to each child were added to the passes to increase the motivating value of engaging with the intervention. Additionally, passes were laminated to keep children from tearing them up during extinction bursts, and the backs of the passes were signed in colored ink by the therapist to preclude forgery.

Rewards for meeting target criteria were selected via a verbal response preference assessment. Daily and weekly rewards were chosen a priori and parents were instructed not to change rewards once selected. Sample daily rewards included a chocolate chip, a lollipop, or access to 10 minutes of screen time. Example weekly rewards included money for a movie on the weekend, a trip to a preferred location in the city, or a pizza party with friends.

Paper-and-pencil tracking forms were provided to the families by the treating therapist. Parents completed these daily and brought them back to each session.

MODIFICATIONS TO THE BEDTIME PASS INTERVENTION

Several modifications were made to Friman and colleagues' (1999) procedures to address the clinical nature of children with anxiety disorders and the concerns of their parents, as well as to target co-sleeping. First, parents were asked at the start of the intervention phase to put co-sleeping on immediate extinction and then to use the bedtime pass procedure to manage expected bedtime resistance behaviors associated with this marked change. Thus, two levels of extinction were part of the intervention—one for co-sleeping and one for all bedtime resistance behaviors engaged in after allotted bedtime passes were used for a given night. Second, the number of bedtime passes each child received per night during the different phases of treatment was determined based on an idiographic assessment rather than having both children only receive a single pass per night during the intervention phase (as was the case in prior studies on the bedtime pass procedure). Given the clinical severity—years spent co-sleeping in parents' beds or staying up all night in attempts to prevent separation—it was suspected that the children might need multiple opportunities per night to seek comfort for their parents to be willing to tolerate the ensuing extinction bursts. And finally,

differential reinforcement of omission of behavior (DRO; see [Vollmer & Iwata, 1992](#), for a review) was added to reward children for staying in their rooms and not using bedtime passes. Differential reinforcement plus extinction procedures may produce less disruption than pure extinction alone ([Lerman et al., 1999](#)). Additionally, an evidence-based approach for treating children with anxiety disorders is exposure and response prevention (ERP; [Ale, McCarthy, Rothschild, & Whiteside, 2015](#)). For ERP to be effective, it is important for an individual to volitionally choose to engage in response prevention rather than being externally blocked from engaging in avoidance or safety behaviors. It was hoped that adding a reinforcement component would incentivize children to choose to face their discomfort by not using the bedtime pass, rather than being blocked from accessing their parents.

OPERATIONAL DEFINITIONS

Independent sleep was defined as the absence of co-sleeping *and* the child staying in their room and not interacting with parents between the end of the nighttime routine and a prespecified wake-up time. If the child briefly left the room independently to get water or use the bathroom without interacting with their parent, that was still considered independent sleep.

Using a bedtime pass was defined as any interaction with parents initiated by the child between the end of the nighttime routine and the prespecified wake-up time. If a child refused to physically hand over the bedtime pass, parents were instructed to consider a pass to have been used and to follow through with planned reactions accordingly.

DESIGN AND ANALYSES

A changing criterion design ([Hartmann & Hall, 1976](#); [Klein, Houlihan, Vincent, & Panahon, 2017](#)) was used for implementation and evaluation of the bedtime pass intervention. Parents collected baseline data to assess independent sleep and frequency of co-sleeping and continued to collect data throughout the intervention phases. The criterion for nights of independent sleep required to earn a prespecified reward was set at the start of each phase. Phase sequences were data driven in that the criterion for earning a reward the following phase was set based on response to the previous phase. Criteria for changing phases were set a priori; however, as the design was implemented in a naturalistic treatment setting, client feedback and scheduling led to phase changes outside of the a priori timeline on a few occasions. The bedtime

pass intervention was assessed to be related to outcomes if the target behavior was seen to vary in response to the changing criterion for earning the reward over the course of treatment. The main outcome variable was the number of nights of independent sleep. Clinically significant change was assessed using visual analysis ([Lane & Gast, 2014](#)).

APPROVAL

Children provided assent for their treatment data to be used for research purposes and their parents also provided consent for the same. Ethical oversight and review of this project was provided by the Behavioral Health Research Collective (BHRC) IRB.

Case Descriptions and Results

Please note that identifiable information about these cases has been modified for de-identification purposes. Both cases illustrate implementation of the bedtime pass protocol, with modifications made consistent with behavioral theory and challenges that inherently arise when conducting treatment in a practice-based setting as opposed to a research trial.

CASE 1: "TALIA"

Talia was a 5-year-old White girl. At the start of treatment her parents were in the process of separating and Talia was primarily living with Parent 1. She visited and stayed the night with Parent 2 on the weekends and one night per week. Talia and Parent 1 presented for assessment and treatment and at the time of the intake, Talia met *Diagnostic and Statistical Manual of Mental Disorders*, 5th edition (DSM-5; [American Psychiatric Association, 2013](#)) criteria for separation anxiety disorder, specific phobias, and oppositional defiant disorder. Talia's separation concerns included intense fears that she or her parents would die, and transitions prompted strong emotional reactions. She also had phobias of bees and thunder and feared for her safety with regard to both cues. She avoided the outdoors despite a strong fondness for parks and nature, scanned the house for over an hour per day looking for insects, and closed windows to protect against weather. In terms of oppositionality, Talia engaged in frequent tantrum behaviors at home. Tantrums typically resulted from being asked to engage in a non-preferred activity, or when told to transition away from a preferred activity. Tantrum behaviors included yelling, crying, falling on the ground, and aggression, with durations of 5 minutes to over an hour per episode. Based on clinical observation, verbal skills were at or above average for a girl her

age. Talia was medication naïve prior to and during treatment.

Despite co-sleeping being identified as problematic by both parents, the initial treatment targets involved treating fear of harm befalling herself or her family, fear of insects, and fear of the natural environment. Given the multifaceted nature of her anxiety and comorbid behavioral problems, a modular CBT approach for treating child anxiety was utilized (Chorpita, 2007). Talia and Parent 1 received developmentally appropriate psychoeducation on anxiety and developed an exposure hierarchy related to her fears. Given the presence of disruptive behaviors, one session was also spent on behavioral parent coaching, teaching functions of behavior and developing an incentive system. Talia then engaged in five sessions of CBT focused on ERP for her anxiety. After these sessions, Talia was able to go outside, leave windows open at home, go into an insect sanctuary, and separate from her parents at a developmentally appropriate level during the day. She no longer reported fear of harm befalling herself or parents at night.

After engaging in CBT for her anxiety symptoms, verbal distress around nighttime routines and getting into bed decreased. However, Talia continued sleeping in her parents' beds despite both parents attempting to transition her into her own room and bed. Attempts to separate from Talia at night resulted in tantrum behavior. Typically, parents would tell her to go sleep in her own room and Talia would protest, stating that it was too hard and too scary. Talia no longer consistently reported specific fears at night, although developmentally appropriate nightmares and fears of sounds at night persisted. Parents consistently acquiesced after tantrums, allowing Talia to sleep in their beds, both due to worries that Talia would not get enough sleep, as well as their own relief from tantrums. Despite a desire for Talia to sleep independently, Parent 1 also initially stated concerns that allowing Talia to "cry it out" (extinction only) would result in a diminished attachment and feared that this would harm their relationship moving forward.

Given the concerns that Parent 1 had about extinction procedures, the bedtime pass intervention was proposed as a strategy to decrease co-sleeping while mitigating the intensity of extinction bursts. Parent 1 found this approach acceptable and agreed to implement it in a graduated manner. Talia was initially ambivalent about decreasing co-sleeping when the bedtime pass intervention was described to her. Treatment began with collection of baseline data.

As can be seen in Figure 1, the Baseline Phase reflected that Talia co-slept with her parents all

night, every night, for 3 weeks in a row. When Talia and Parent 1 presented for treatment the next week, time was spent discussing the intervention and increasing motivation for engagement with the intervention by discussing potential reinforcers with Talia. Idiographic bedtime passes were created in session with Talia's favorite pictures on them. She stated high motivation to keep a pass by staying in her room for the night. Procedures for the plan were practiced in the session using a doll in the place of Talia, and Parent 1 practiced walking the doll back to its bed after the doll turned in the bedtime pass and received brief attention. It was also made clear to Talia and Parent 1 that starting in Phase One, co-sleeping would be placed on extinction. Talia's parents decided for treatment and logistical reasons that for the first month of the intervention, Talia would visit with Parent 2, but sleep every night at Parent 1's house, with plans to generalize procedures to Parent 2's house once proficiency was demonstrated.

In Phase One (Week 1), Talia was provided with seven bedtime passes to keep in her room (one for each night of the week). Once Parent 1 said goodnight and they finished Talia's bedtime routine, Talia was expected to stay in her room, without calling out, until 6:00 A.M. If during the night she wanted parent attention, comfort, or to ask a question, she would turn in a bedtime pass and then would have up to 15 minutes of attention. At the end of 15 minutes, Parent 1 would walk Talia back to her room. If Talia again sought attention in the same night, Parent 1 would ignore her behaviors and silently physically guide Talia back to her room. Every night Talia engaged in independent sleep and did not turn in a bedtime pass, she would receive a few chocolate chips with breakfast the following morning. Additionally, it was specified that during the first week of the intervention, should Talia keep at least two bedtime passes (i.e., meet the criteria for independent sleep at least two nights), she would earn a trip to the local aquarium.

Talia was unexpectedly unable to return to treatment the following week due to illness, but she and Parent 1 returned to treatment after Week 2. Parent 1 reported that at the end of the first week, Talia had kept two bedtime passes, earning a trip to the aquarium. Parent 1 said that on the nights that Talia turned in bedtime passes and received attention, Talia needed moderate redirection back to her bed, with Talia coming out of the room multiple times asking questions or requesting that Parent 1 get her water. Parent 1 kept the criterion for earning a reward consistent during Week 2 (a tea party with Parent 1 for keeping two passes).

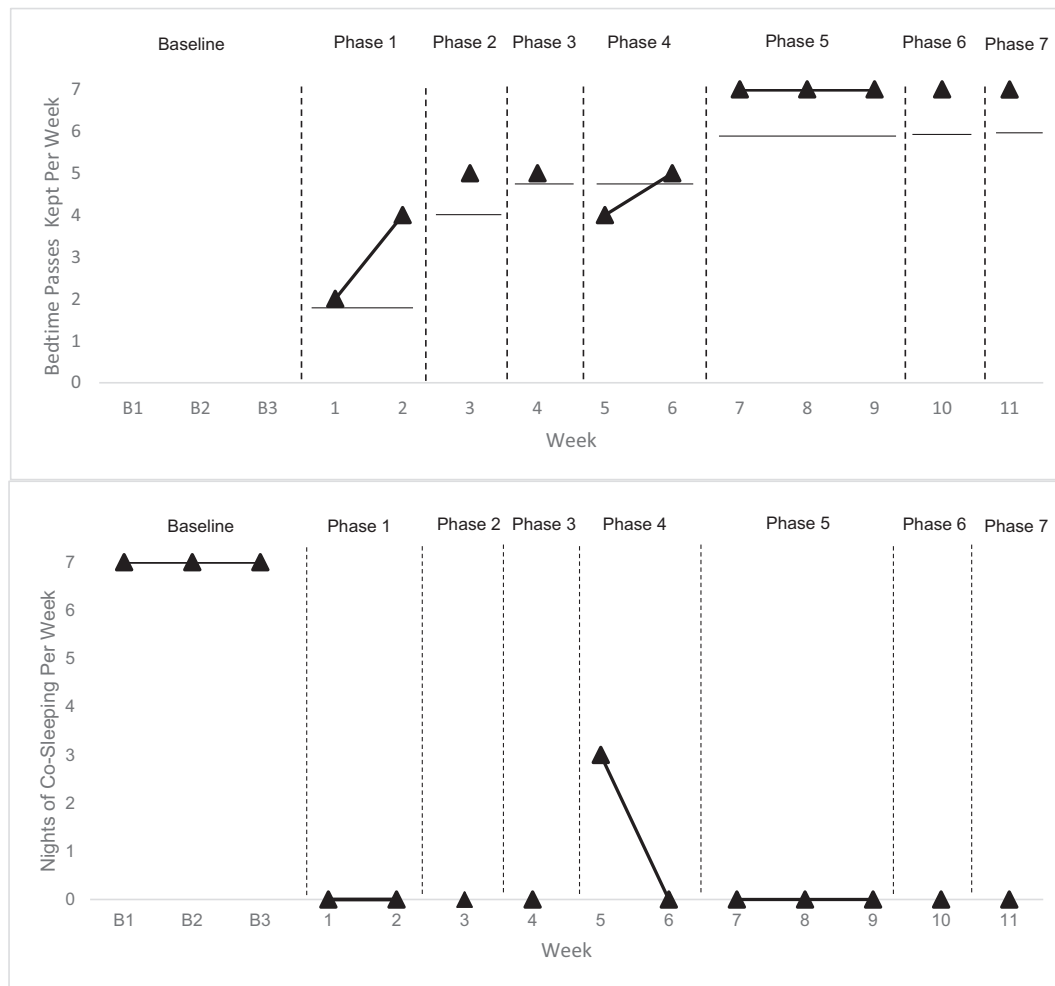


FIGURE 1 Talia's behavioral response to the bedtime pass intervention. (Top) The number of bedtime passes Talia kept per week and the criterion threshold for earning a reward. The number of bedtime passes kept per week is the same as the number of nights of independent sleep as she only had one pass per night throughout the intervention. There is no data on bedtime passes during the baseline period, as she had not yet been given passes. (Bottom) The number of nights Talia engaged in co-sleeping per week.

During Week 2, Talia surpassed the criterion by retaining bedtime passes four out of seven nights. Given the reported success, Phase Two was implemented, with the criterion for earning an end-of-the-week reward increased to keeping four bedtime passes. The following week (Week 4), Phase Three was implemented with the criterion for earning an end of the week reward increased to keeping five bedtime passes. As can be seen in Figure 1, Talia met the criteria for independent sleep five out of seven nights during Phases Two and Three. At the end of the third phase, Parent 1 reported that on the nights when Talia was using her bedtime pass, she no longer needed redirection or came out of her room once she had used the one pass available to her.

It was decided that Phase Four (Week 5) would focus on generalizing the intervention to Parent 2's

house. The intervention was explained to Parent 2 over the phone, and they were given time to ask questions. They demonstrated proficiency and understanding of the intervention by answering the therapist's questions. It was decided that Talia would sleep at Parent 2's house three nights per week, once during the school week and twice on the weekend. A criterion of five passes was set in order to earn a weekly reward, thus necessitating that Talia meet criteria for independent sleep at Parent 2's house at least once in the week. The weekly reward was to be delivered by Parent 2. Talia continued to earn small pieces of chocolate at breakfast following each night that she did not turn in a bedtime pass.

When Talia returned to treatment at the end of Week 5, it was reported that she had not met the threshold for the weekly reward; though she met

the criteria all four nights at Parent 1's house, these behaviors did not generalize to Parent 2's house. Parent 2 reported that they had trouble with the extinction-based aspects of the intervention. After Talia turned in her bedtime pass, Parent 2 allowed Talia to co-sleep for the rest of the night and this occurred all three nights at Parent 2's house. The importance of extinction was reiterated. The daily reward earned in the morning at Parent 2's house was modified based on Talia's preference for a trinket rather than a chocolate chip. As Talia had not met the criterion of five nights of independent sleep, she did not earn the weekly reward and the criterion remained the same the next week.

At the end of Week 6, Talia had met criteria for independent sleep four nights at Parent 1's house one night at Parent 2's house. Parent 2 reported that they had implemented extinction procedures for co-sleeping and for any behaviors following the use of a bedtime pass. She earned her weekly reward. Phase Five lasted the next 3 weeks, with a criterion of six nights of independent sleep established for earning a weekly reward. At this point, Talia began to sleep in her room, by herself, throughout the night, seven nights per week, while continuing her normal schedule of sleeping at both parent's homes. For Phase Six (Week 10), it was established that the daily reward would be faded out for independent sleep, and the six-bedtime pass threshold for the weekly reward would remain. Talia again met criteria for independent sleep seven nights. In the final phase, Phase Seven (Week 11), weekly rewards were faded into a more "natural intervention" with Talia earning a small monetary allowance if she met criteria for independent sleep at least six of seven nights of the week. Talia again met this criteria all seven nights.

Following completion of work on co-sleeping, Talia remained in treatment to address avoidant reactions to loud noises and her parents also received additional behavioral parent coaching for oppositional behaviors. At 3 and 6 months after Phase Seven of the bedtime pass intervention, she continued to sleep in her own bed on a nightly basis (i.e., no co-sleeping), with only developmentally appropriate intermittent requests for parent attention at night.

CASE 2: "LIAM"

Liam was a 9-year-old White boy living with both of his parents. He and both parents presented for assessment and treatment for excessive fear of the dark, so much so that he had trouble going to different parts of the home alone at night and co-slept with his parents every night. Liam reported specific worries, such as concerns about monsters

and robbers, as well as a more general discomfort with the dark. He requested that all lights be left on at night and that parents provide excessive reassurance that they would not turn off the lights. Both Liam and his parents stated a desire that he sleep independently in his own bed and room. Liam presented with what appeared to be average or above average verbal skills and was enrolled in advanced classes at school. At intake, Liam met DSM-5 (APA, 2013) criteria for a specific phobia of the dark. He was later diagnosed with GAD. Liam was medication naïve prior to and during treatment.

The primary target of treatment was co-sleeping, as Liam and his parents stated that bed sharing was significantly interfering with family functioning. Parents began collecting data on frequency of independent sleep after the intake. Notably, his parents initially reported hesitation to use extinction as a treatment component for fear this would damage their relationships with Liam. In contrast to Talia's parents, Liam's parents requested that treatment begin with differential reinforcement without extinction. While this deviated from the specified bedtime pass procedure, it allowed for a test of the efficacy of the bedtime pass without extinction for co-sleeping. See Figure 2 for Liam's behavioral performance in each phase of treatment.

As can be seen in Figure 2, Baseline Phase (1 week) data reflected that Liam co-slept with his parents 100% of nights. Parents were able to leave his room about half an hour after the onset of sleep; however, if Liam awoke in the middle of the night, he would go find his parents and require that they come back and sleep in his bed. When Liam and his parents presented for treatment the next week, the bedtime pass intervention was explained and Liam brainstormed potential reinforcers. Bedtime passes were created in session with Liam's favorite pictures on them. He stated high motivation to keep a pass by staying alone in his room for the night. Procedures for the plan were practiced in session using the therapist in the place of Liam, and Liam's parents practiced ignoring the therapist until he turned in a bedtime pass. Because Liam's parents were still hesitant to engage in extinction components of treatment, it was agreed that Liam would begin the night in his room alone, but should he turn in a bedtime pass, a parent would sleep in his bed for the rest of the night. Additionally, Liam and his parents reported that he experienced generalized worries throughout the day (about his schedule, grades, friendships) that resulted in physical fatigue, and it was agreed that in addition to the bedtime pass intervention, Liam would engage in concurrent CBT for anxiety symptoms, as described below.

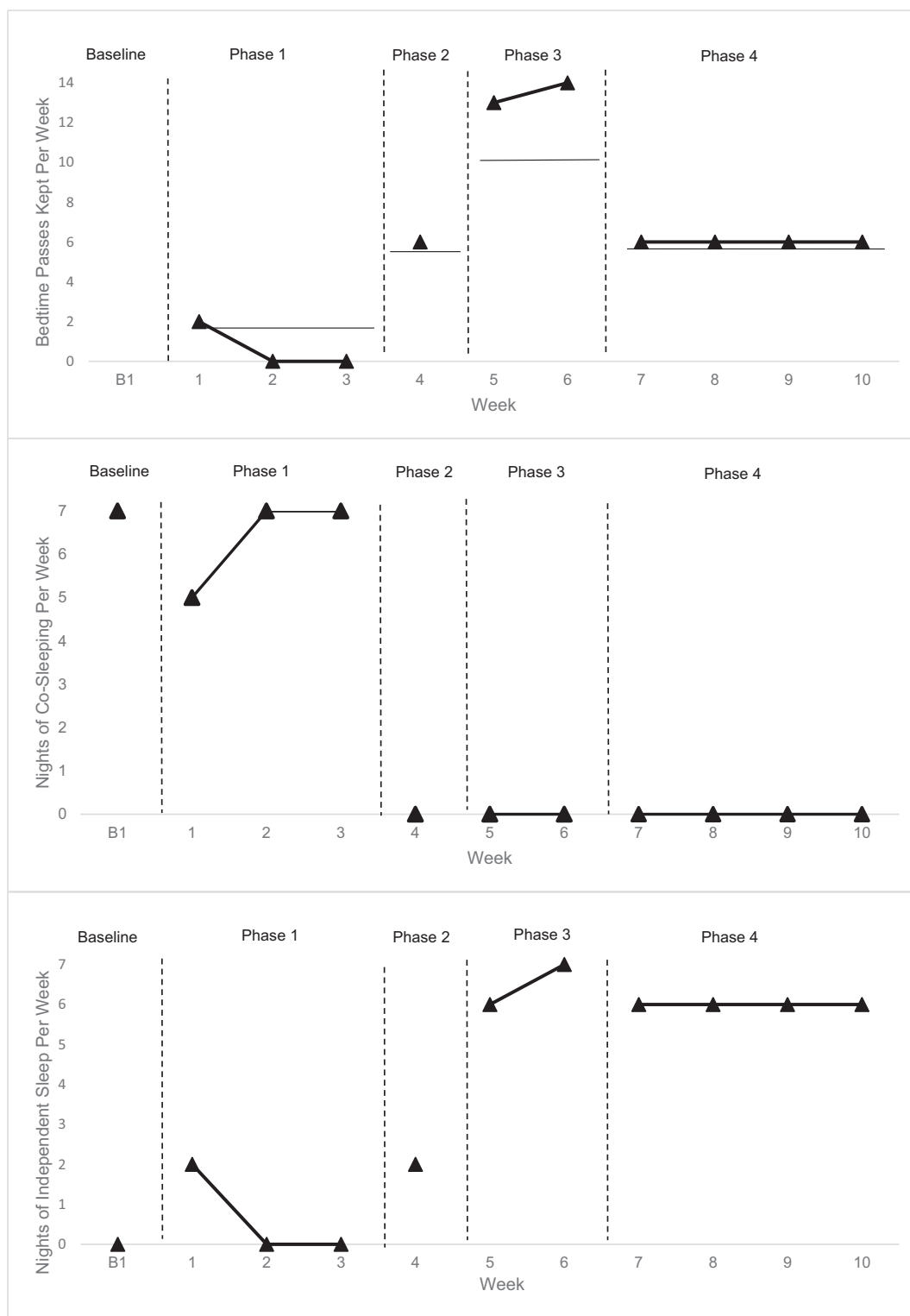


FIGURE 2 Liam's behavioral response to the bedtime pass intervention. (Top) The number of bedtime passes Liam kept per week and the criterion threshold for earning a reward. There is no data on bedtime passes during the baseline period, as he had not yet been given passes. (Middle) The number of nights Liam engaged in co-sleeping per week. (Bottom) The number of nights per week that Liam engaged in independent sleep. As Liam had access to two bedtime passes per night for some phases of the intervention, this is represented separately from the number of bedtime passes kept per week.

At the start of Phase One (Week 1 after baseline), Liam was given seven bedtime passes to keep in his room (one per night) and was offered a movie night with his parents if he kept two bedtime passes. Additionally, if Liam kept a bedtime pass at night, 10 minutes of additional screen time were available the next morning. As seen in Figure 2, when Liam and his parents returned, Liam had slept alone in his own room and not used any bedtime passes two nights. Liam was incredibly proud, as these were the first times he had gone without co-sleeping.

While *a priori* procedures specified a phase change with an increase in the criterion for earning a reward, Liam's parents were hesitant and said that they would not comply. They said that on the two nights that Liam met criteria for independent sleep, he demonstrated high levels of anxiety and made verbal statements about not feeling prepared to increase criteria for earning rewards. Despite the therapist's reservations, the phase and criterion remained the same, with two nights of not co-sleeping and meeting criteria for independent sleep required for earning a reward. Additionally, progressive muscle relaxation (PMR) skills (Chorpita, 2007) were introduced and practiced in session as a coping strategy Liam could use when anxious.

Despite his success during the first week of the intervention, Liam did not meet criteria for independent sleep at all during Week 2, choosing to have access to parents and forgoing the available rewards. At the start of Week 3, Liam chose what he saw as a very potent reward and the criterion for number of nights of independent sleep to earn the reward remained the same (two). The focus of anxiety treatment that week centered on education about the role of avoidance and parent accommodation in maintaining anxiety over time. Despite statements about increased motivation, Liam again did not have any nights of independent sleep during Week 3. The focus of Session 4 was on increasing Liam's parents' motivation to incorporate extinction into the bedtime pass procedure. They reiterated fear about harming their relationship with Liam but stated a great desire for him to stop co-sleeping. After validating the emotion behind their concerns, motivational strategies were used to shift their willingness to engage in this component of treatment and by the end of the session they agreed to put co-sleeping on immediate extinction and to fully implement the bedtime pass plan (also with extinction). Liam was taught guided imagery to feel as if he continued to develop coping tools for his anxiety.

Phase Two started in Week 4 and the bedtime pass procedure (with extinction and differential

reinforcement) was implemented. Liam started *each* night of the week with two bedtime passes. The additional pass per night (vs. the usual one pass per night) was provided based on baseline data that Liam was leaving his bed to find his parents multiple times per night should they leave his bed for any sustained period of time. According to the new procedures, once Liam's parents said good-night and they finished Liam's bedtime routine, Liam was expected to stay in his room without calling out until 5:30 A.M. or later. If during the night he wanted parent attention, comfort, or to ask a question, he would turn in a bedtime pass, and then would have up to five minutes of attention. At the end of the 5 minutes, a parent would walk Liam back to his room. If Liam again sought attention during the same night, he could turn in one more bedtime pass and receive up to 5 more minutes of attention. Once he had used both of his passes and been guided back to his room, parents were instructed to ignore his subsequent behaviors and silently physically guide Liam back into his room. For every bedtime pass that Liam kept at night he earned 5 minutes of additional screen time the next day. Additionally, it was specified that during Phase Two of the intervention, should Liam keep at least six bedtime passes for the week, he would earn a fun trip with his friends on the weekend (reward selected via verbal preference assessment). Note that during this phase, number of nights of independent sleep was not set as a criterion. Rather, criterion for the weekly reward was established based on the number of passes that Liam maintained (as he had access to more than one pass per night).

Liam and his parents returned to treatment at the start of Week 5 reporting that Liam slept in his own bed, without co-sleeping, every night of the previous week with no instances of upset. Additionally, he kept 6 of 14 passes. Two nights Liam met the criteria for independent sleep by not interacting with his parents at all, two of the nights he turned in a single pass for 5 minutes of attention, and three of the nights he turned in both passes for a cumulative 10 minutes of attention. Liam's parents were especially excited about their perception that their relationship had not been negatively impacted by the extinction procedures and were more than willing to increase the criterion for earning a reward. In Phase Three of treatment (Weeks 5 and 6) the criterion for earning a weekly reward was increased to 10 bedtime passes maintained per week (thus requiring at least three nights of meeting criteria for independent sleep).

At the end of Week 6, Liam and his parents returned to treatment reporting that he surpassed

criterion of keeping 10 bedtime passes both weeks. In Week 5, Liam used a single bedtime pass, keeping 13 by the end of the week (meeting criteria for independent sleep six nights of the week) and earning an outing with friends. In Week 6, Liam kept all his bedtime passes, meeting criteria for independent sleep all seven nights of the week, earning a trip to a doughnut shop. Additionally, Liam independently began entering dark rooms without his parents and turning off his lights at night, and discontinued asking for reassurance as to where his parents would be without being instructed to do so in treatment. Notably, no formal exposure sessions had yet taken place as part of CBT for his anxiety, although undoubtedly Liam was engaging in incidental exposure by remaining in his room at night without accessing his parents.

For the final phase of treatment, Phase Four, the number of bedtime passes given to Liam was faded and he was provided with one pass per night. He could earn up to 10 minutes of extra screen time per day should he keep the pass from the evening before. A criterion of six passes maintained per week was established for earning his weekly reward, or he could bank the reward and earn a larger reward at the end of the month. Liam's parents were to implement this plan for the next 4 weeks before the next treatment session. This length of time was due to logistical reasons outside of the context of the treatment plan and could not be modified.

Liam and his parents returned to treatment after 4 weeks and said that Liam had slept in his own bed, without co-sleeping, every night over the previous 4 weeks. He turned in one bedtime pass for brief parental attention each week since the prior session (meaning he met criteria for independent sleep six nights out of seven each week). Anecdotally, Liam would typically turn in a pass when he felt worried about the outcome of events of the next day and wanted to discuss his anxiety (e.g., if he had a spelling test the next day he would turn in a pass and ask for reassurance about his readiness for the test). Despite the presence of some mild anxiety, Liam and his parents were thrilled with the progress of treatment and felt that the bedtime pass intervention could continue to be implemented in a similar fashion in perpetuity. At the time they saw the intervention as highly acceptable and saw no need for further treatment.

Around 2 years after discontinuing treatment, Liam's parents contacted the clinician. They said that they had faithfully implemented the bedtime pass intervention for 2 months after ending treatment and that Liam was sleeping in his own

bed, without co-sleeping, all nights. However, after 2 months the family took a vacation where they all slept in the same room for over 2 weeks. Upon returning home, Liam demonstrated a return of fear, with great difficulty separating from his parents and fear of sleeping alone for fear that kidnappers would break into the house and harm him. Liam's parents said that although they found the bedtime pass intervention to be acceptable with regards to maintaining the relationship with their son, they had not been willing to exert the emotional effort needed to implement extinction-based procedures again. They continued to make rewards available for meeting the criteria of independent sleep, but resumed allowing co-sleeping (i.e., took co-sleeping off extinction). When Liam's parents reached out, they said that Liam had been co-sleeping with them every night for about 2 years.

While this return to baseline (or baseline plus DRO) was not planned, it provided a test of the active component of the bedtime pass intervention for Liam. He did not meet criteria for independent sleep during the periods (Baseline Phase, Phase One, the nearly 2 years after relapse of co-sleeping) when his parents implemented differential reinforcement (bedtime pass reward) without extinction. He was fully independent up to the criterion set when extinction procedures were implemented. Around two years after the discontinuation of the initial course of treatment, Liam resumed care with the same therapist. After he received eight sessions of CBT for GAD, including ERP and one session dedicated to behavioral parent instruction, parents were again willing to implement extinction-based methods using the bedtime pass procedure and Liam once again stopped co-sleeping and consistently met criteria for independent sleep.

Discussion

The cases presented demonstrate the potential of using a modified bedtime pass procedure to help children with anxiety disorders and their parents reduce problematic co-sleeping and manage related bedtime resistance. The data from these cases extend prior research on the use of the bedtime pass intervention with nonclinical children (Freeman, 2006; Friman et al., 1999; Moore et al., 2007) to children with anxiety disorders and targeting co-sleeping specifically. These data also address an important gap in the literature given the high rates of co-sleeping in children and adolescents with anxiety disorders (Alfano et al., 2007; Palmer et al., 2018; Storch et al., 2008) and the findings that after effective treatment for anxiety symptoms, many anxious youth continue to co-sleep and

struggle with sleeping independently (Alfano et al., 2007; Storch et al., 2008). The bedtime pass intervention is relatively straightforward to implement and could have widespread utility for anxious children and their families who are struggling to fade co-sleeping behaviors, especially given that parents in the current cases, and in prior research (Friman et al., 1999), viewed it more favorably than extinction-only procedures.

Talia and Liam were both co-sleeping with their parents 100% of nights at the start of treatment. The bedtime pass procedure was introduced to Talia and her parents following CBT for her anxiety symptoms. She, like many children who undergo successful anxiety treatment (Peterman et al., 2016), was still experiencing residual sleep problems. Liam's case represents use of the bedtime pass procedure with a child who had not first had a full course of CBT for anxiety; although he was given psychoeducation and introduced to PMR and guided imagery concurrently with the bedtime pass intervention, his anxiety symptoms were not directly targeted with ERP. Both Talia and Liam fully discontinued co-sleeping and only sought occasional developmentally appropriate attention from parents at night by the end of treatment. Their behavior changes followed criterion established for earning weekly rewards, but only when their parents implemented extinction-based procedures. It is notable that Liam's fear and avoidance of the dark decreased markedly with decreases in co-sleeping and increases in nighttime independence. This is hypothesized to be due to naturalistic exposure to being alone at night.

Several modifications to the bedtime pass procedure were made. Consistent with prior research (Etherton et al., 2016), neither Talia nor Liam's parents were willing to use extinction-only methods to reduce co-sleeping, despite the considerable evidence supporting the efficacy of this approach (Etherton et al., 2016; Mindell et al., 2006). Therefore, the bedtime pass procedure was selected to pair with putting co-sleeping on extinction, as it has been shown to help manage bedtime resistance behaviors that were expected to increase once extinction was put into place. Thus, the first change was putting co-sleeping on immediate extinction. The second change was that bedtime passes were not limited to one per night, as in prior studies. Instead, the number of passes per night was based on idiographic data about the children's behaviors and then faded, if needed, down to one per night by the end of treatment. Although Talia received one pass per night throughout treatment, Liam's number of passes per night was increased from one to two when his parents were willing to place co-sleeping on

extinction. This change allowed for more of a tailored intervention given that Liam frequently left his room when his parents were not present during baseline. The final modification was the inclusion of differential reinforcement, both daily and weekly, to reward absence of bedtime resistance and encourage children to stay in their rooms throughout the night (and retain bedtime passes). It is possible that the bedtime pass procedure alone (as used in prior research) would have been enough to help transition these children to independent sleep, but given the clinical nature of the sample and the common practice of including differential reinforcement in child anxiety treatment (Chorpita, 2007), rewards were included. Future research could examine whether differential reinforcement is an essential component of the intervention when used to target co-sleeping in anxious children.

It is notable that several obstacles were encountered in the attempt to implement the intervention as planned, all of which pertained to the use of extinction procedures. For Talia, Parent 1 was willing to place co-sleeping on immediate extinction and bedtime passes were used to mitigate anticipated bedtime resistance related to this change. Despite around 4 weeks of no co-sleeping at Parent 1's house, when Talia transitioned to Parent 2's house and co-sleeping was initially not put on extinction (a deviation from the intended procedures), she chose to co-sleep every night this was allowed. When co-sleeping was put on extinction at Parent 2's home, she again stopped co-sleeping and used bedtime passes as needed. Liam's parents were initially unwilling to put co-sleeping on extinction and instead wanted to try reducing co-sleeping using only the bedtime pass procedure (including differential reinforcement for retaining passes). Despite initially being able to not co-sleep for two nights and being motivated to earn rewards for not co-sleeping and retaining passes, Liam was unable to consistently stop co-sleeping until his parents placed co-sleeping on extinction (Phase Two). The data from both cases are consistent with that from Freeman (2006), who found that use of the bedtime pass without extinction was less effective and optimal results were only achieved when extinction was included. Despite these families facing barriers to extinction procedures similar to those reported by others (e.g., difficulties tolerating crying, fears about how extended distress would impact their bond, methods being inconsistent with their personal beliefs; Etherton et al., 2016), extinction was successful once implemented. Anecdotally, at the end of treatment all parents reported feeling that the intervention was acceptable and that the bond with their child was intact.

After treatment was over and Liam's co-sleeping behaviors resumed and his nighttime fear increased, his parents were not willing to go through the emotional challenge of re-implementing the bedtime pass intervention (with extinction). It is possible that parent-only interventions, motivational enhancement, or planned follow-up sessions would have helped them maintain their gains. This hypothesis is supported in that when they resumed treatment two years later, they reinstated the procedures and Liam was able to stop co-sleeping again. However, another factor that must be considered is that Liam did not initially engage in formal ERP for his anxiety symptoms as Talia had (as his fear at night decreased while treating his co-sleeping). It is possible that this left him vulnerable to return of fear, especially with the contextual shift of being on vacation with his parents for two weeks and the family sleeping in the same room during that time. Future research could examine the durability of the bedtime pass procedure as a stand-alone intervention for co-sleeping and bedtime resistance for children with anxiety, as well as in combination with gold-standard interventions like CBT for anxiety (Higa-McMillan et al., 2016), to see which leads to better maintenance of gains.

It is important to highlight that despite some deviations from intended protocols and challenges with implementing extinction, these families demonstrated many strengths that likely facilitated the success of the intervention. For instance, one of Talia's parents suspended overnight custodial visits for a month to allow sufficient time to implement the intervention in one parent's home before generalizing it to their home. Additionally, all parents showed a willingness to make efforts to modify nighttime behaviors, risking potentially more sleep disruption in the short-term, despite everyone needing to go to work and school most mornings. These kinds of strengths are often seen when working with families in clinical settings, but it is important to note that in families with less willingness, the intervention may be less effective.

Several limitations of the current procedures and data are important to note. It would have been a stronger test of the efficacy of the intervention had there been a phase with a decrease in the criterion for earning the reward (Klein et al., 2017), thereby demonstrating the behavior conformed to the criterion rather than maintaining a continuing trajectory. This was not deemed clinically feasible or acceptable, a notable limitation of conducting research in a practice setting. However, phase lengths were varied and there were at least three phases of the intervention for each child, and the children's behavioral responses in each phase

adhered relatively well to the established criterion. On several occasions, behaviors surpassed the requirement to earn the weekly reward, decreasing absolute confidence that behavior varied with the established criterion. It should also be noted that on a couple of occasions there were unintended deviations from the established procedures, such as when Talia's Parent 2 did not place co-sleeping on extinction or when Liam's parents withdrew the intervention after a vacation. While these allowed for a glimpse at important components of the intervention, the unintended nature of the modifications may have weakened the experimental design and limited the generalizability of the findings. Another limitation is the small number of cases reported on and the limited range of anxiety disorders present (GAD, separation anxiety disorder, specific phobias). Larger case series and randomized controlled trials that include children with the full range of anxiety and related disorders would help extend generalizability. Future research could also work to identify which children with anxiety and sleep problems benefit from adjunctive interventions, like the bedtime pass procedure, and which demonstrate remission of SRPs in response to CBT for anxiety symptoms alone. This would provide clinicians with guidance for clinical decision making when treating children with co-morbid anxiety and sleep problems.

Despite limitations, a major strength of the current data is that they were collected as part of naturalistic treatment and the procedures used, and modifications made in response to clinical issues and parental concerns, are reflective of the challenges that clinicians face when implementing evidence-based practices. Incorporating a changing criterion design into the treatment plan also allowed for a rigorous evaluation of the effectiveness of the intervention and helped to identify potential essential components for these children (e.g., extinction). Overall, these data provide preliminary support for the use of a modified bedtime pass procedure with anxious children who co-sleep, which, to our knowledge, is the first such data published. The intervention described here can be used as a stand-alone intervention, or as an adjunct to standard CBT for pediatric anxiety. Also, the highly flexible nature of the intervention makes it easy to tailor to the specific needs and contexts of individual children and their families. Such variability is illustrated in the cases presented (i.e., different number of bedtime passes per night to start, different family living situations, different types of rewards). The procedure may appeal to a much wider range of parents than extinction-only procedures, perhaps making it more likely to be

implemented and successful. More research is needed to further substantiate the current findings. However, we hope that the methods described here are useful for clinicians treating children with comorbid anxiety and co-sleeping, given the gaps in the literature on this topic. We also hope that these cases can serve as additional examples of how clinicians can contribute to the empirical literature by taking interventions with research support and modifying them to address clinical problems that do not yet have an established evidence-base for treatment.

Conflict of Interest Statement

The authors declare that there are no conflicts of interest.

References

- Ale, C. M., McCarthy, D.M., Rothschild, L. M., & Whiteside, S. P. (2015). Components of cognitive behavioral therapy related to outcome in childhood anxiety disorders. *Clinical Child and Family Psychology Review*, 18, 240–251. <https://doi.org/10.1007/s10567-015-0184-8>
- Alfano, C. A., Ginsburg, G. S., & Kingery, J. N. (2007). Sleep-related problems among children and adolescents with anxiety disorders. *Journal of the American Academy of Child & Adolescent Psychiatry*, 46, 224–232. <https://doi.org/10.1097/01.chi.0000242233.06011.8e>
- Alfano, C. A., Pina, A. A., Zerr, A. A., & Villalta, I. K. (2010). Pre-sleep arousal and sleep problems of anxiety-disordered youth. *Child Psychiatry and Human Development*, 41, 156–167. <https://doi.org/10.1007/s10578-009-0158-5>
- American Psychiatric Association (2013). *Diagnostic and statistical manual of mental disorders* (5th ed.). Washington, DC: APA. <https://doi.org/10.1176/appi.books.9780890425596>
- Chase, R. M., & Pincus, D. B. (2011). Sleep-related problems in children and adolescents with anxiety disorders. *Behavioral Sleep Medicine*, 9, 224–236. <https://doi.org/10.1080/15402002.2011.606768>
- Chorney, D. B., Detweiler, M. F., Morris, T. L., & Kuhn, B. R. (2008). The interplay of sleep disturbance, anxiety, and depression in children. *Journal of Pediatric Psychology*, 33, 339–348. <https://doi.org/10.1080/15402002.2011.606768>
- Chorpita, B. F. (2007). *Modular cognitive-behavioral therapy for childhood anxiety disorders*. Guilford Press. <https://doi.org/10.1177/1049731507302268>
- Clementi, M., & Alfano, C. (2014). Targeted behavioral therapy for childhood generalized anxiety disorder: A time-series analysis of changes in anxiety and sleep. *Journal of Anxiety Disorders*, 28, 215–222. <https://doi.org/10.1016/j.janxdis.2013.10.006>
- Etherton, H., Blunden, S., & Hauck, Y. (2016). Discussion of extinction-based behavioral sleep interventions for young children and reasons why parents may find them difficult. *Journal of Clinical Sleep Medicine*, 12, 1535–1543. <https://doi.org/10.5664/jcsm.6284>
- Freeman, K. A. (2006). Treating bedtime resistance with the bedtime pass: A systematic replication and component analysis with 3-year-olds. *Journal of Applied Behavior Analysis*, 39, 423–428. <https://doi.org/10.1901/jaba.2006.34-05>
- Friman, P. C., Hoff, K. E., Schnoes, C., Freeman, K., Woods, D. W., & Blum, N. (1999). The bedtime pass: An approach to bedtime crying and leaving the room. *Archives of Pediatrics and Adolescent Medicine*, 153, 1027–1029. <https://doi.org/10.1001/archpedi.153.10.1027>
- Hartmann, D. P., & Hall, R. V. (1976). The changing criterion design. *Journal of Applied Behavior Analysis*, 9, 527–532. <https://doi.org/10.1901/jaba.1976.9-527>
- Higa-McMillan, C. K., Francis, S. E., Rith-Najarian, L., & Chorpita, B. F. (2016). Evidence base update: 50 years of research on treatment for child and adolescent anxiety. *Journal of Clinical Child & Adolescent Psychology*, 45, 91–113. <https://doi.org/10.1080/15374416.2015.1046177>
- Klein, L. A., Houlihan, D., Vincent, J. L., & Panahon, C. J. (2017). Best practices in utilizing the changing criterion design. *Behavior Analysis in Practice*, 10, 52–61. <https://doi.org/10.1007/s40617-014-0036-x>
- Lane, J.D., & Gast, D. L. (2014). Visual analysis in single case experimental design studies: Brief review and guidelines. *Neuropsychological Rehabilitation*, 24, 445–446. <https://doi.org/10.1080/09602011.2013.815636>
- Lerman, D. C., Iwata, B.A., & Wallace, M.D. (1999). Side effects of extinction: Prevalence of bursting and aggression during the treatment of self-injurious behavior. *Journal of Applied Behavior Analysis*, 32, 1–8. <https://doi.org/10.1901/jaba.1999.32-1>
- Mileva-Seitz, V. R., Bakermans-Kranenburg, M. J., Battaini, C., & Luijk, M. P. C. M. (2017). Parent-child bed-sharing: The good, the bad, and the burden of evidence. *Sleep Medicine Reviews*, 32, 4–27. <https://doi.org/10.1016/j.smrv.2016.03.003>
- Mindell, J. A., Kuhn, B., Lewin, D. S., Meltzer, L. J., & Sadeh, A. (2006). Behavioral treatment of bedtime problems and night wakings in infants and young children. *Sleep*, 29, 1263–1276. <https://doi.org/10.1093/sleep/29.10.1263>
- Moore, B.A., Friman, P. C., Fruzzetti, A. E., & MacAleese, K. (2007). Brief report: Evaluating the bedtime pass program for child resistance to bedtime - A randomized, controlled trial. *Journal of Pediatric Psychology*, 32, 283–287. <https://doi.org/10.1093/jpepsy/jsl025>
- Palmer, C. A., Clementi, M.A., Meers, J. M., & Alfano, C. A. (2018). Co-sleeping among school-aged anxious and non-anxious children: Associations with sleep variability and timing. *Journal of Abnormal Child Psychology*, 46, 1321–1332. <https://doi.org/10.1007/s10802-017-0387-1>
- Peterman, J. S., Carper, M. M., Elkins, R. M., Comer, J. S., Pincus, D. B., & Kendall, P. C. (2016). The effects of cognitive-behavioral therapy for youth anxiety on sleep problems. *Journal of Anxiety Disorders*, 37, 78–88. <https://doi.org/10.1016/j.janxdis.2015.11.006>
- Peterman, J. S., Carper, M. M., & Kendall, P. C. (2015). Anxiety disorders and comorbid sleep problems in school-aged youth: Review and future research directions. *Child Psychiatry and Human Development*, 46, 376–392. <https://doi.org/10.1007/s10578-014-0478-y>
- Storch, E. A., Murphy, T. K., Lack, C. W., Geffken, G. R., Jacob, M. L., & Goodman, W. K. (2008). Sleep-related problems in pediatric obsessive-compulsive disorder. *Journal of Anxiety Disorders*, 22, 877–885. <https://doi.org/10.1016/j.janxdis.2007.09.003>
- Tate, R. L., Perdices, M., Rosenkoetter, U., McDonald, S., Togher, L., Shadish, W., ... Vohra, S. (2016). The single-case reporting guideline in behavioral interventions (SCRIBE) 2016: Explanation and elaboration. *Archives of*

- Scientific Psychology*, 4, 10–31. <https://doi.org/10.1037/arc0000026>
- Thompson-Hollands, J., Kerns, C. E., Pincus, D. B., & Comer, J. S. (2014). Parental accommodation of child anxiety and related symptoms: range, impact, and correlates. *Journal of Anxiety Disorders*, 28, 765–773. <https://doi.org/10.1016/j.janxdis.2014.09.007>
- Vollmer, T. R., & Iwata, B.A. (1992). Differential reinforcement as treatment for behavior disorders: Procedural and functional variations. *Research in Developmental Disabilities*, 13, 393–417. [https://doi.org/10.1016/0891-4222\(92\)90013-V](https://doi.org/10.1016/0891-4222(92)90013-V)

RECEIVED: September 16, 2019

ACCEPTED: March 19, 2020

AVAILABLE ONLINE: 24 March 2020